

THE UNIVERSITY OF ZAMBIA
UNIVERSITY SECOND SEMESTER EXAMINATIONS - NOV/DEC, 1996

DE 112

ELEMENTS OF STATISTICAL DEMOGRAPHY
IN SOCIO-ECONOMIC PLANNING

TIME: THREE HOURS

INSTRUCTIONS: ATTEMPT ANY FOUR QUESTIONS. ALL QUESTIONS
CARRY EQUAL MARKS.

1. The formulation, adoption and implementation of a housing policy in Zambia is topical. Discuss.
2. (a) Out of 5,000 university students enrolled in one academic year only 3,500 were enrolled the following academic year.
 - (i) Calculate the drop-out rate
 - (ii) Calculate the progression rate(b) 1,000 Doctors attended to 15,000 women, 3,000 children and 2,000 men.
 - (i) Calculate the Doctor/patient ratio.
 - (ii) How relevant is this ratio in planning?
3. (a) What comprises the social security package? and
(b) how can social security be measured?
4. (a) What type of financial statistics are available and how can they be used in Zambia.
5. Categorise the socio-economic indicators that are usable in Zambia.

END OF EXAMINATION

THE UNIVERSITY OF ZAMBIA

UNIVERSITY SECOND SEMESTER EXAMINATIONS - NOV/DEC 1996

DE 312

COMPONENTS OF POPULATION CHANGE

TIME: THREE HOURS

INSTRUCTIONS: ANSWER ANY THREE (3) QUESTIONS

- Q1. a) Zambia is one of the countries with high fertility. What factors have led to such patterns of fertility?
- b) What type of policies or programmes would you ~~adopt~~ to curb the increase in fertility.
- Q2. a) Briefly discuss the determinants of the following in Zambia:
1. Fertility increase
 2. Mortality increase
- b) What are the likely consequences of the 2 components in section a.
- Q3. Describe the likely consequences of rural to urban migration in a developing country like Zambia.
- Q4. a) Discuss the present differentials in mortality and fertility between less developed and more developed regions in the world today.
- b) What are some of the causes of such differences in mortality and fertility.
- Q5. a) Discuss the patterns of international migration in Africa today.
- b) Describe what type of policies programmes that can be adopted to curb or encourage or control international migration in any country
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END OF EXAMINATION

THE UNIVERSITY OF ZAMBIA

UNIVERSITY EXAMINATIONS - NOVEMBER/DECEMBER 1996

DE 322

INTRODUCTION TO DEMOGRAPHIC ANALYSIS

TIME: THREE HOURS

INSTRUCTIONS: ALL CANDIDATES MUST ANSWER QUESTION 1 AND ANY OTHER TWO QUESTIONS FROM QUESTIONS 2, 3, AND 4 FOR A TOTAL OF THREE QUESTIONS.

ANSWER QUESTION 1

1. Briefly discuss the concepts and measurements below, paying particular attention to their merits and demerits as well as the assumptions underlying each:
- (a) Crude birth/death rate
 - (b) Child-woman ratio
 - (c) General fertility rate
 - (d) Age specific death/birth rate
 - (e) Standardized death/birth rate
 - (f) Life expectancy
 - (g) The concept of person years

ANSWER ANY TWO QUESTIONS FROM QUESTIONS 2, 3, AND 4.

2. (a) Discuss the differences between specific indirect and direct methods of measuring migration. What are their underlying assumptions? Under what circumstances is one preferred over the other? Any strengths and/or weaknesses of the methods?
- (b) Give the formulas for the following basic measurements of migration and, for each, explain what it attempts to measure:
- (i) The in-migration rate
 - (ii) The out-migration rate
 - (iii) The gross-migration rate
 - (iii) The net-migration rate
- (c) Given below are data from Zambia. Use the national growth rate method to estimate migration and comment on

the results. Do you see any interesting results?

	1980	1990
KABWE	136033	161455
LUANSHYA	129589	141927
CHIPATA	204738	291372
MANSA	111437	142497
LUSAKA	535830	769353
CHINSALI	93999	83659
KASAMA	113935	189360
SOLWEZI	92773	124624
LIVINGSTONE	71521	82952
MONGU	114402	142795
ZAMBIA	5661801	7383097

3.
 - (a) Briefly explain what a life table is.
 - (b) Give some important uses of a life table.
 - (c) In life table construction, it is always important to convert the central death rate, M_x , into a probability of dying, q_x . What logic underlies this? (Credit will be given for giving the conversion formula)
 - (d) A generation of 10,000 ants was born on the 1/1/1985. 7,000 were alive on 1/1/1986, 5000 on 1/1/1987, 2,000 on 1/1/1988, 500 on 1/1/1989 and the whole generation was dead on 1/1/1990. Construct a life table from the above information with the following functions:
 l_x d_x p_x L_x M_x T_x and e_x
 (Hint: $M_x = d_x / L_x$)
 - (e) What is the birth rate in the population represented by this life table of ants? What is the death rate?
 - (f) What is the probability of an ant aged 2 celebrating its 4th birthday
4.
 - (a) What is the difference between an age specific rate and parity by age.
 - (b) What is the difference between completed family size and total fertility rate. In what way are they similar?
 - (c) Use the data below to compute the following:
 - (i) The total fertility rate
 - (ii) The gross reproduction rate
 - (iii) The net reproduction rate
 - (e) For all the three measurements of fertility above, give their advantages and disadvantages as well as their underlying assumptions.
 - (d) On the basis of your computations, would you say this population is replacing itself or not. Briefly explain.

AGE	NUMBER OF WOMEN	NUMBER OF BIRTHS	FEMALE SURVIVAL RATES
15-19	480162	38537	.97417
20-24	428007	100210	.97131
25-29	381755	94573	.96827
30-34	308741	62822	.96438
35-39	279640	43192	.95866
40-44	211469	14745	.95000
45-49	187480	4240	.93642

END OF EXAMINATION